

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 35-2025: A 5-Year-Old Boy with Fever and Rash

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PRESENTATION OF CASE

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CME



Dr. Ann M. Murray: A 5-year-old boy was admitted to this hospital because of fever and rash.

The patient had been well until 3 days before the current presentation, when his mother noticed that he was not eating and drinking as much as usual. Later that day, the oral temperature measured at home was 39.2°C. The patient's mother gave him acetaminophen, and his fever abated.

Two days before the current presentation, an itchy rash appeared on the dorsal surface of both feet and spread toward the shins. The patient's mother took him to the emergency department of another hospital.

On evaluation, the temporal temperature was 36.7°C. The patient appeared well. Diffuse blanching macules and papules were present on the legs, and a few scattered papules were seen on the buttocks and lower back. The rash was considered to be consistent with exposure to poison ivy. Treatment with ibuprofen was administered, a course of oral prednisone was prescribed, and the patient was discharged home. The patient's mother was instructed to give him antihistamines as needed for itching.

One day before the current presentation, the patient's mother gave him diphenhydramine and applied calamine lotion to his body; she had not filled the prescription for prednisone. The patient seemed to scratch the rash less often; however, it spread to his arms. The patient's mother took him to the pediatric primary care clinic affiliated with the other hospital.

On examination, the temporal temperature was 37.0°C, and the patient appeared well. The conjunctivae were injected, and there was a palpable posterior cervical lymph node on the right side that was soft and mobile. Small, circular, dark red-purple, nonblanching papules were present on the legs, buttocks, and lower back, and papules had begun forming on the forearms. The blood levels of electrolytes, alanine aminotransferase, aspartate aminotransferase, and total bilirubin were normal, as were the results of tests of kidney function. Other laboratory test results are shown in Table 1. The patient's mother was instructed to monitor the

rash at home, and a follow-up appointment was scheduled.

When the patient awoke on the day of the current presentation, he was scratching his skin continuously and refused to walk. The oral temperature measured at home was 40.2°C and did not decrease after acetaminophen and ibuprofen were given. The patient's mother brought him to the emergency department of this hospital.

On presentation, the patient had no sore throat, abdominal pain, nausea, vomiting, or diarrhea. His medical history included iron deficiency, constipation, reactive airway disease, and benign sleep myoclonus. He had met normal growth and developmental milestones and had received all routine childhood vaccinations. Medications included albuterol as needed for wheezing, diphenhydramine and calamine lotion as needed for itching, and acetaminophen and ibuprofen as needed for fever. He had no known adverse reactions to medications. The patient lived with his parents in a suburban area of New England, and until 5 days before this admission, he had been attending a local summer camp, where he participated in various outdoor activities. He had no history of travel outside the region. His mother had anemia, and his father had hypertension.

The temporal temperature was 38.1°C, the blood pressure 98/69 mm Hg, the heart rate 102 beats per minute, the respiratory rate 26 breaths per minute, and the oxygen saturation 100% while the patient was breathing ambient air. The patient appeared uncomfortable and was tearful. The conjunctivae were injected, and the right tympanic membrane was erythematous. Erythematous macules were seen on the palate and buccal mucosa (Fig. 1A), with no ulceration or fissuring of the lips. Palpable posterior cervical lymphadenopathy was present on the right side. The lungs were clear on auscultation, and the abdomen was nontender. He had nonblanching papules on the legs (Fig. 1B), buttocks, lower back, arms, and ears, with overlying excoriations. The hands and feet were swollen. He had mild erythema but no rash on the palms and soles. Laboratory test results are shown in Table 1.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Mitchell J. Feldman: This fully immunized 5-year-old boy with asthma presented with 2 days of fever and a pruritic rash that had begun on the legs and had spread to involve the buttocks, lower back, arms, mouth, and ears. The rash initially

Table 1. Laboratory Data.

Variable	Reference Range, Other Hospital	1 Day before Current Presentation, Other Hospital	Reference Range, This Hospital [*]	On Presentation, This Hospital
White-cell count (per μ l)	6000–17,000	9680	5500–15,500	15,850
Hematocrit (%)	33.0–39.0	35.4	34.0–40.0	37.8
Hemoglobin (g/dl)	10.2–13.5	11.8	11.5–13.5	12.4
Platelet count (per μ l)	130,000–400,000	211,000	150,000–450,000	247,000
Differential count (per μ l)				
Neutrophils	1800–7000	6070	1500–8500	13,090
Lymphocytes	1000–4000	1950	2000–8100	1240
Monocytes	0–800	770	200–1700	820
Eosinophils	0–450	750	0–1200	550
Basophils	0–200	80	0–500	140
Erythrocyte sedimentation rate (mm/hr)	0–10	20	0–14	35
C-reactive protein (mg/liter)	0.1–3.0	16.7	0–8.0	20.0

* Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges shown are those used at Massachusetts General Hospital for children who do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

included both macules and papules, which then evolved to red-purple, nonblanching papules consistent with purpuric lesions. The patient had swelling of the hands and feet, and he appeared uncomfortable and refused to walk. The combination of fever and rash in a child is a commonly encountered clinical presentation. In working toward the most likely diagnosis in this patient, I will consider the distinctive morphologic features, distribution, and evolution of the rash, in addition to the mucosal lesions, conjunctival injection, lymphadenopathy, elevated levels of inflammatory markers, and possible recent exposures.

BACTERIAL INFECTION

In a child with fever and a purpuric rash, it is urgent that meningococemia be ruled out. Given the patient's young age, he would not yet have received meningococcal vaccine, which is usually administered at 11 years of age in the United States. The typical rash that appears in patients with meningococcal infection is petechial; however, rash with macular and urticarial morphologic features may also be seen, with a

truncal and distal distribution that usually spares the head, palms, and soles.¹ Purpura fulminans, which can occur with meningococemia and disseminated intravascular coagulation, is characterized by skin findings that are more intensely purple and larger than those described in this patient. In addition, the consumptive coagulopathy associated with purpura fulminans would be expected to result in thrombocytopenia, which was absent in this patient.² Furthermore, the 3-day course of this patient's illness would be atypical for meningococcal disease, which more commonly develops over a period of hours.

Rocky Mountain spotted fever is an important consideration in this child who recently participated in outdoor activities at a local summer camp. He had no known tick bites; however, previous bites are reported in only about half of cases of tickborne illness.³ Rocky Mountain spotted fever is prevalent in the southeastern United States but may occur in New England, especially on Cape Cod and Martha's Vineyard.⁴ Although the rash associated with Rocky Mountain spotted fever begins peripherally and spreads

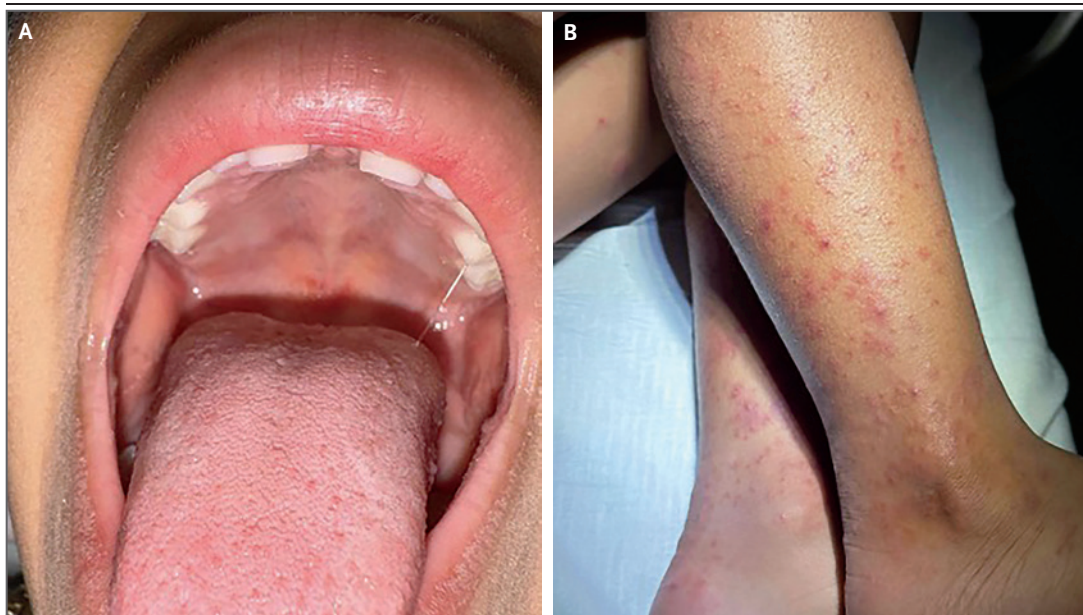


Figure 1. Clinical Photographs.

On examination in the emergency department, conjunctival erythema was present in both eyes (not shown); erythematous macules are noted on the palate and buccal mucosa, without ulceration or fissuring of the lips (Panel A); and a nonblanching, raised, papular rash is present on the legs (Panel B), buttocks, lower back, arms, and ears, with overlying excoriations.

centripetally, it typically develops several days after onset of fever, whereas this patient's rash occurred the next day. In addition, headache, abdominal pain, vomiting, and thrombocytopenia, which are common findings in patients with Rocky Mountain spotted fever, were absent in this patient.

VIRAL INFECTION

Despite the recent resurgence of measles cases in the United States, this would be an unlikely diagnosis in this patient because he was fully immunized. Conjunctival injection was present, but he had no antecedent symptoms of upper respiratory infection. The rash associated with measles begins on the face and spreads toward the body and legs — unlike the progression of the rash in this patient. In addition, the macular erythematous lesions seen in his mouth were not consistent with the Koplik spots found in patients with measles, which typically have a more white or gray appearance.

Papular–purpuric “gloves and socks” syndrome is characterized by acral edema, pruritic rash, and fever and is associated with several viral infections, including parvovirus B19 infection, Epstein–Barr virus infection, and cytomegalovirus infection. This syndrome is most common in adolescents and young adults but has been reported in young children. Mucosal lesions may be present but vesiculopustular lesions are more commonly seen,⁵ which is inconsistent with the macular erythematous lesions described in this patient. Furthermore, the rash associated with papular–purpuric gloves and socks syndrome does not typically extend proximal to the wrists and ankles.^{6,7}

The Gianotti–Crosti syndrome, the eponym for papular acrodermatitis of childhood, is also associated with viral infections and occurs most commonly in young children. The rash is usually papular, and the distribution can be similar to that observed in this patient, including the extensor surfaces of the arms and legs, the buttocks, the face, and the ears.⁸ Purpura may be present. Although patients with the Gianotti–Crosti syndrome are usually asymptomatic, a mildly pruritic rash may develop in some patients, and the presence of a fever may reflect the antecedent viral trigger.

Multisystem inflammatory disease in children

(MIS-C) can complicate severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection in rare cases. We are not provided with information about recent SARS-CoV-2 infection or exposure, but the patient had many of the common presenting symptoms of MIS-C, including fever, rash, conjunctivitis, lymphadenopathy, and swollen hands and feet. Neutrophilia and elevated levels of inflammatory markers were also present. However, since the patient had no evidence of cardiac, hematologic, or gastrointestinal involvement, the findings in this case do not meet the proposed diagnostic criteria for MIS-C.⁹

Hand, foot, and mouth disease is caused by enteroviruses such as coxsackievirus A16 and enterovirus A71, and it most commonly occurs during the summer. The associated fever is usually mild, and the typical rash is papulovesicular.¹⁰ Although purpuric rash can develop, skin lesions are usually nonpruritic, unlike the rash in this patient. In addition, the patient's oral lesions were described as erythematous macules, as opposed to vesicles and aphthae, which are characteristic findings of hand, foot, and mouth disease.

This patient does not have a travel history that would be consistent with dengue fever or Ebola virus disease, which are worthy of consideration in a patient with fever and purpura. Neither of these conditions are acquired locally in Massachusetts.^{11,12}

FUNGAL INFECTION

Disseminated histoplasmosis may lead to the development of multiple erythematous papules and oral lesions. However, this condition typically occurs in patients with severe immunocompromise and is uncommon in the northeastern United States.¹³

DRUG REACTION

Drug reaction with eosinophilia and systemic symptoms (DRESS), the Stevens–Johnson syndrome, and hypersensitivity vasculitis can all manifest as erythematous maculopapular and purpuric eruptions. However, all these conditions are precipitated by the use of medication, typically antibiotic agents and antiepileptic drugs. Before the onset of fever and rash, this patient's only medication was albuterol, which is not commonly associated with these conditions.¹⁴

GENETIC DISEASE

Familial Mediterranean fever is an autosomal recessive hereditary disease caused by a variant in *MEFV*. This disease usually manifests in childhood with recurrent fever. Rashes on the arms and legs may be present in 13% of cases but are typically erysipelas-like, which is not consistent with the features of this patient's rash.¹⁵

CANCER

Purpura, fever, lymphadenopathy, and refusal to walk can be clues to a diagnosis of leukemia. In a study reviewing the presentations of 203 children with acute lymphoblastic leukemia, fever was present in 60% of the patients, lymphadenopathy in 57%, and purpura in 30%.¹⁶ However, the occurrence of purpura in the context of this condition is associated with thrombocytopenia, which was absent in this patient. In addition, no hepatomegaly or splenomegaly was noted on examination, and the complete blood count and differential count showed no evidence of blasts.¹⁷

PHYSICAL CHILD ABUSE

Because bruises, which can have an appearance similar to that of purpura, were present in this patient in locations including the buttocks and ears in the context of a normal platelet count, physical child abuse should also be considered. Although high fevers at home were reported, the temperature was normal during evaluation at the other hospital and at the primary care office and was only mildly elevated on presentation to this hospital. This suggests the possibility of factitious fever, which may also be associated with child abuse. However, this patient's skin lesions are not consistent with nonaccidental trauma, such as an imprint of a hand or a pattern from an object used to strike.^{18,19}

VASCULITIS

Kawasaki's disease is a common form of vasculitis seen in children in the United States. It should be considered early in the evaluation of patients with fever and rash, since prompt treatment with intravenous immune globulin can prevent the occurrence of coronary artery aneurysms reported in up to 25% of untreated patients.²⁰ This patient presented with all five of the classic examination findings that are part of the diagnostic criteria for Kawasaki's disease:

conjunctival injection, intraoral macular erythema, edema of the hands and feet, maculopapular rash, and cervical adenopathy (although it was not reported whether the lymph nodes were ≥ 1.5 cm in diameter).²⁰ The patient was tearful and appeared uncomfortable, features suggestive of irritability, which is often seen in patients with this condition. However, he did not have at least 4 to 5 days of fever, which is required for even a diagnosis of incomplete Kawasaki's disease.²⁰ Furthermore, this patient's rash evolved into purpura, which is atypical of the rash associated with Kawasaki's disease. However, given that the patient's presentation meets many of the criteria for the diagnosis of Kawasaki's disease and that the fever pattern may still be emerging, I would obtain an echocardiogram to rule out coronary artery aneurysms.²⁰⁻²²

IGA VASCULITIS

IgA vasculitis (also known as Henoch-Schönlein purpura) is the most common vasculitis of childhood, and this patient's age matches the peak incidence of 4 to 6 years.²³ IgA vasculitis may have various triggers, including infections and medications. In patients with this condition, deposits of IgA immune complexes, specifically the IgA1 subclass, are found in the small vessels of the skin, gastrointestinal tract, and renal glomeruli. The hallmark of IgA vasculitis is nonthrombocytopenic palpable purpura on the legs, with the classic tetrad also including arthralgia or arthritis, abdominal pain, and kidney disease (Fig. 2). The diagnostic criteria require that patients have purpura (usually palpable and appearing in crops) or petechiae, with predominance in the lower limbs, and that such findings are not related to thrombocytopenia. In addition, patients must have at least one of the following features: abdominal pain, histopathological confirmation, arthritis or arthralgia, or kidney involvement.²⁴

There was no description of abnormal joint findings in this case, but the patient's refusal to walk on the day of admission could indicate arthralgia. He had no history of abdominal pain, but only 50% of children with IgA vasculitis have gastrointestinal symptoms. We are not provided with the results of a urinalysis; hematuria is the most common abnormality seen in patients who have IgA vasculitis with kidney involvement. Angioedema of the arms or legs, which was

present in this patient, has been described in 46% of cases,²⁵ and elevated levels of inflammatory markers are common.

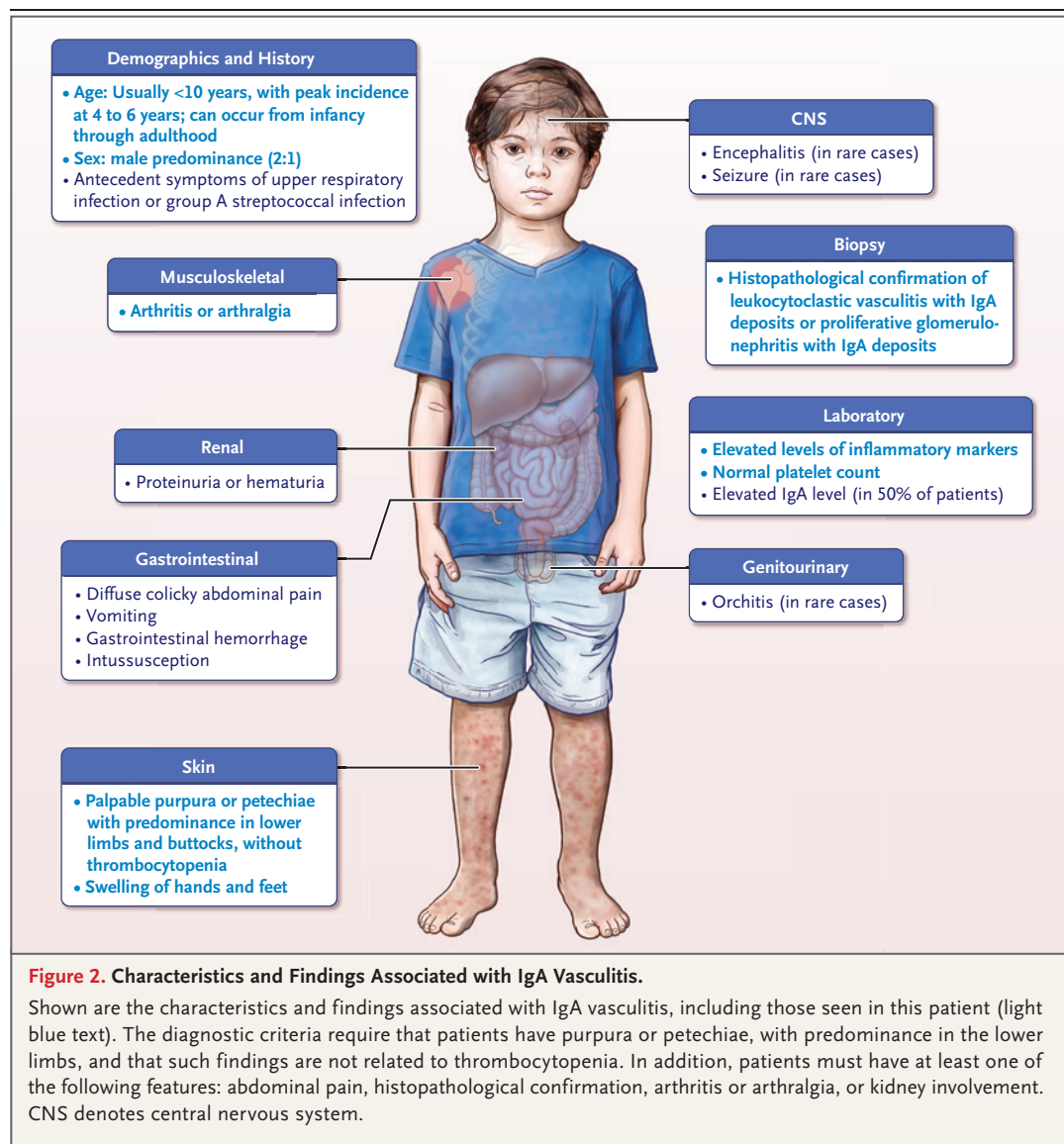
Several features of this patient's presentation — including fever, conjunctival injection, oral erythema, and cervical lymphadenopathy — are uncommon findings in patients with IgA vasculitis but have been described.²⁵⁻³¹ The rash associated with IgA vasculitis is typically not pruritic or only mildly pruritic. Perhaps the pruritic nature of this patient's rash could be explained by his history of reactive airway disease and the known association between atopy and pruritus.

On the basis of the case description, the mor-

phologic features of the skin lesions, the classic distribution and evolution of the rash, and the ability to explain the most important findings, I suspect that the most likely diagnosis in this patient is IgA vasculitis and that the diagnostic test was a skin biopsy showing leukocytoclastic vasculitis with IgA deposits.^{24,32-34} I would also recommend performing echocardiography given the possibility of evolving Kawasaki's disease.

DR. MITCHELL J. FELDMAN'S DIAGNOSIS

IgA vasculitis.



CLINICAL IMPRESSION

Dr. Murray: Although IgA vasculitis was our working diagnosis for this patient at the time of consultation with the infectious diseases service, some features of his presentation were different from the typical IgA vasculitis presentation. He had a severely pruritic rash that was more dramatic than the rashes seen in other cases of IgA vasculitis. The fact that the patient presented during the summer was also atypical; the peak incidence of IgA vasculitis in children occurs during the fall and winter months, when upper respiratory infections, which often trigger IgA vasculitis, are more prevalent.³⁵ Viral infections identified as triggers for IgA vasculitis include those caused by parainfluenza virus, influenza viruses, respiratory syncytial virus, SARS-CoV-2, adenovirus, parvovirus B19, and Epstein-Barr virus.^{36,37} Group A streptococcal infection is an important bacterial trigger in cases of IgA vasculitis.³⁵ *Helicobacter pylori* infection and *Mycoplasma pneumoniae* infection are other less common bacterial triggers associated with IgA vasculitis.

Given the severity of this patient's presentation and the presence of atypical features, the dermatology service was consulted and a skin biopsy was performed to confirm the suspected diagnosis of IgA vasculitis. Echocardiography was also performed to evaluate the possibility of Kawasaki's disease complicated by coronary artery aneurysm. In the context of recent outdoor exposure at a summer camp, high fever, conjunctivitis, a rash that started on the feet and ankles, swelling of the feet, and possible arthralgia, empirical treatment for Rocky Mountain spotted fever with doxycycline was initiated while further diagnostic evaluation was pending.

Dr. Elena B. Hawryluk: At the time of the dermatology consultation, the evolution of the patient's cutaneous findings was reviewed. The influence of skin pigmentation on the appearance of the rash was also taken into consideration (i.e., the skin phototype on the Fitzpatrick scale, which ranges from type I [very light skin that always burns and never tans] to type VI [very dark skin that never burns, tans very easily, and is deeply pigmented]). In lighter skin types, bright red erythematous papules contrast with skin; in darker skin types, erythema may appear

more violaceous, as was the case for this patient with Fitzpatrick skin type IV. One helpful technique is to apply pressure (diascopy) to examine for blanching, which can show the effect of vasculature on skin appearance.

This patient's rash initially manifested as blanching pink macules. However, the appearance of the macules evolved over the course of his illness. They had become darker and non-blanching at the time of presentation to this hospital, and later, overlying papules and vesicles developed. Over a period of days, the vesicles appeared hemorrhagic and dusky, and pink macules darkened to red nonblanching macules, coalescing to patches in dependent areas.

From a dermatologic perspective, the patient's presentation was most suggestive of leukocytoclastic vasculitis. Skin biopsy is not required for diagnosis in patients with typical presentations of vasculitis. However, in patients with atypical presentations, such as this patient, skin biopsy can help to confirm the diagnosis and an immunofluorescence assay can define the subtype of vasculitis, which can guide management decisions. When skin biopsy is performed, it is important to test skin lesions that have been present for less than 24 hours and to provide separate samples for hematoxylin and eosin staining and for immunofluorescence.

PATHOLOGICAL DISCUSSION

Dr. Miekia A. Stonhill: A punch biopsy of the skin on the right lateral thigh was performed. Hematoxylin and eosin staining of sections of the biopsy specimen revealed a prominent dermal perivascular neutrophilic infiltrate with areas of leukocytoclasia, along with fibrinoid necrosis of vessel walls and extravasation of erythrocytes (Fig. 3). No pathologically significant dermal edema was present. Taken together, these findings were diagnostic of leukocytoclastic vasculitis. An additional punch-biopsy specimen was obtained for direct immunofluorescence testing, which showed focal vascular deposition of IgA and C3, findings that support the clinical diagnosis of IgA vasculitis.

PATHOLOGICAL DIAGNOSIS

IgA vasculitis.

DISCUSSION OF MANAGEMENT

Dr. Holly Rothermel: Patients presenting with IgA vasculitis are often acutely ill and have palpable purpura accompanied by clinical findings that include joint pain, abdominal pain, and glomerulonephritis. It is often treated in the outpatient setting and requires hospitalization only if patients have severe symptoms.^{38,39} The acute phase of IgA vasculitis resolves in most children within 1 month, although patients can have recurrent symptoms in the first year after diagnosis. Hydration, pain control, and antipyretic medications are the mainstays of treatment.⁴⁰ Nonsteroidal antiinflammatory medications such as naproxen and ibuprofen are effective for joint and abdominal pain and are not contraindicated unless a child has glomerulonephritis or gastrointestinal bleeding.⁴⁰

Although most autoimmune types of vasculitis are treated with glucocorticoids and immunosuppressive therapy, this is not the case in patients with mild-to-moderate IgA vasculitis without kidney involvement. Glucocorticoids are used to treat joint pain, rash, orchitis, and abdominal pain, but data do not support the early use of glucocorticoid therapy for the prevention of kidney involvement.⁴¹⁻⁴³ Most clinicians reserve glucocorticoid treatment for children who are hospitalized with severe symptoms. When glucocorticoids are deemed necessary, intravenous administration is often needed, since gut edema can decrease the absorption of therapy.⁴⁴ When glucocorticoids are used, the dose is generally tapered over the course of 3 to 4 weeks to avoid rebound symptoms.

Long-term renal disease is the most serious complication that can result from IgA vasculitis; monitoring urinalyses during the first 6 months of disease is critical. Although proteinuria or hematuria occurs in up to one third of patients, clinically important kidney disease develops in less than 10% of children with IgA vasculitis.⁴⁵ Although the findings were not included in the initial case history provided to Dr. Feldman, this patient had no evidence of hematuria or proteinuria. In one study involving more than 1000 children with IgA vasculitis, no long-term renal impairment was observed among patients who, like this patient, had a normal urinalysis on presentation, and less than 2% of the patients

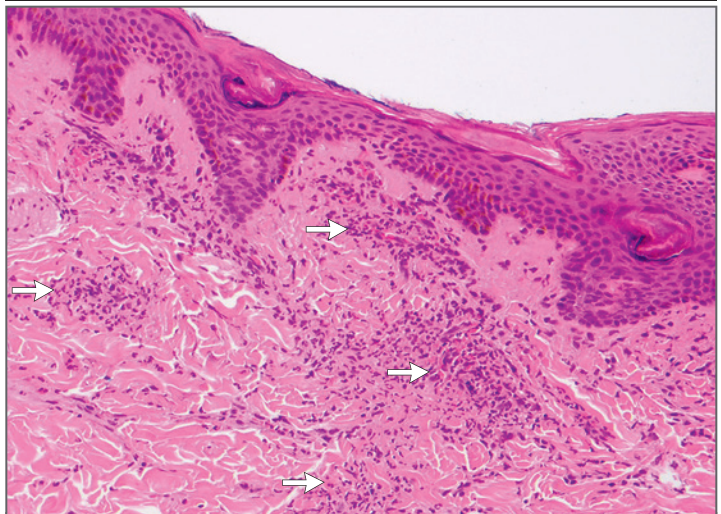


Figure 3. Skin-Biopsy Specimen.

Hematoxylin and eosin staining of a punch biopsy specimen obtained from the skin of the right lateral thigh shows features consistent with small-vessel vasculitis (arrows), which is characterized by fibrinoid necrosis of the vascular wall and perivascular infiltrate of neutrophils, with leukocytoclasia and red-cell extravasation.

with isolated hematuria or mild proteinuria had ongoing kidney disease. In contrast, long-term kidney disease developed in almost 20% of the patients with nephrotic-range proteinuria, elevated creatinine levels, and hypertension.⁴⁵ If this patient had had moderate or severe proteinuria or decreased kidney function, a kidney biopsy would have been recommended to evaluate for severe kidney disease, which is usually treated with more extensive immunosuppressive therapy.^{40,46,47}

FOLLOW-UP

Dr. Rothermel: This patient received presumptive treatment with oral prednisone for possible vasculitis and doxycycline for possible infection. His fever resolved after 1 day, but the rash worsened. Treatment with doxycycline was stopped after the skin biopsy confirmed the diagnosis of IgA vasculitis. On hospital day 4, facial edema, abdominal pain, leg pain, and scrotal edema developed. A repeat urinalysis showed no hematuria or proteinuria. Treatment with intravenous methylprednisolone was initiated for his worsening pain and edema, and his symptoms abated. He was discharged with a prescription for a

3-week tapering course of oral glucocorticoids. At a follow-up visit 1 week later, the patient had no pain, and the skin lesions were healing. A repeat urinalysis was performed 6 months after discharge and showed no evidence of glomerulonephritis.

FINAL DIAGNOSIS

IgA vasculitis.

This case was presented at the Harvard Medical School 2024 postgraduate course “Primary Care Pediatrics,” directed by Peter

T. Greenspan, Ronni L. Goldsmith, Benjamin A. Nelson, John Patrick T. Co, and Janice A. Lowe and administered by Jennifer Timulty.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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